LAB FAT

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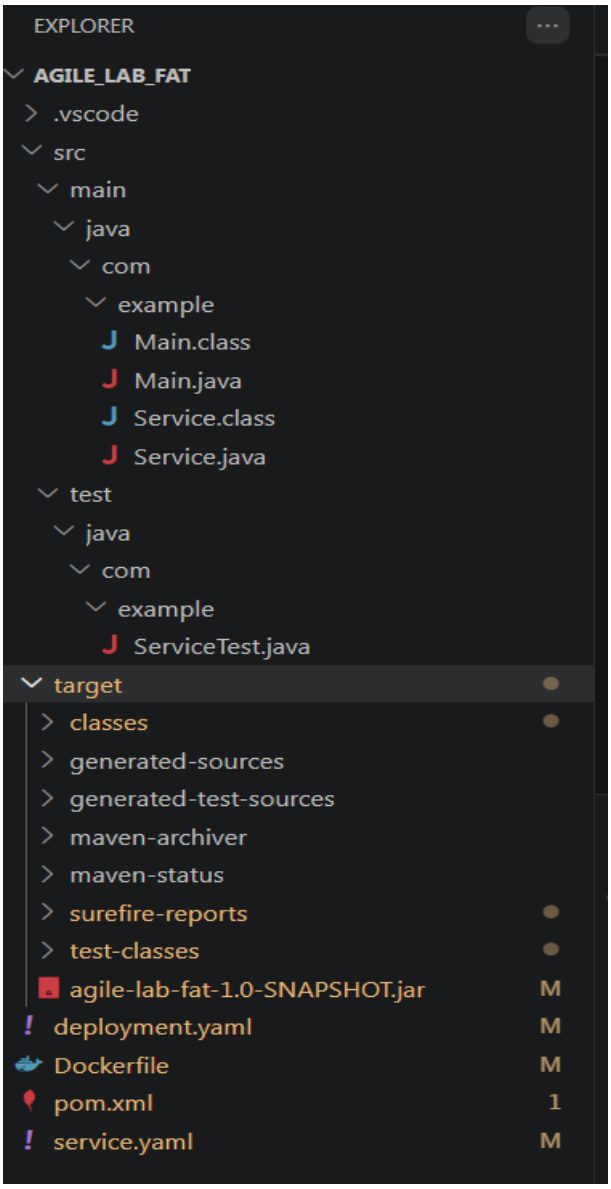
Course Name: Agile Development Process and DevOps (LAB)

Course Code: ISWE406P

Slot: L3+L4

Step 1: Create the project folder

```
AGILE_LAB_FAT
├── src
│   ├── main
│   │   └── java
│   │       └── com
│   │           └── example
│   │               ├── Main.java
│   │               └── Service.java
│   └── test
│       └── java
│           └── com
│               └── example
│                   └── ServiceTest.java
└── pom.xml
```



Step 2: Write Java Code

Main.java

```
package com.example;

public class Main {

    public static void main(String[] args) {

        Service service = new Service();

        String result = service.resetPassword("1234", "Anusha-agile");

        System.out.println(result);
        System.out.println("Updated Password: " +
service.getCurrentPassword());
    }
}
```

Service.java

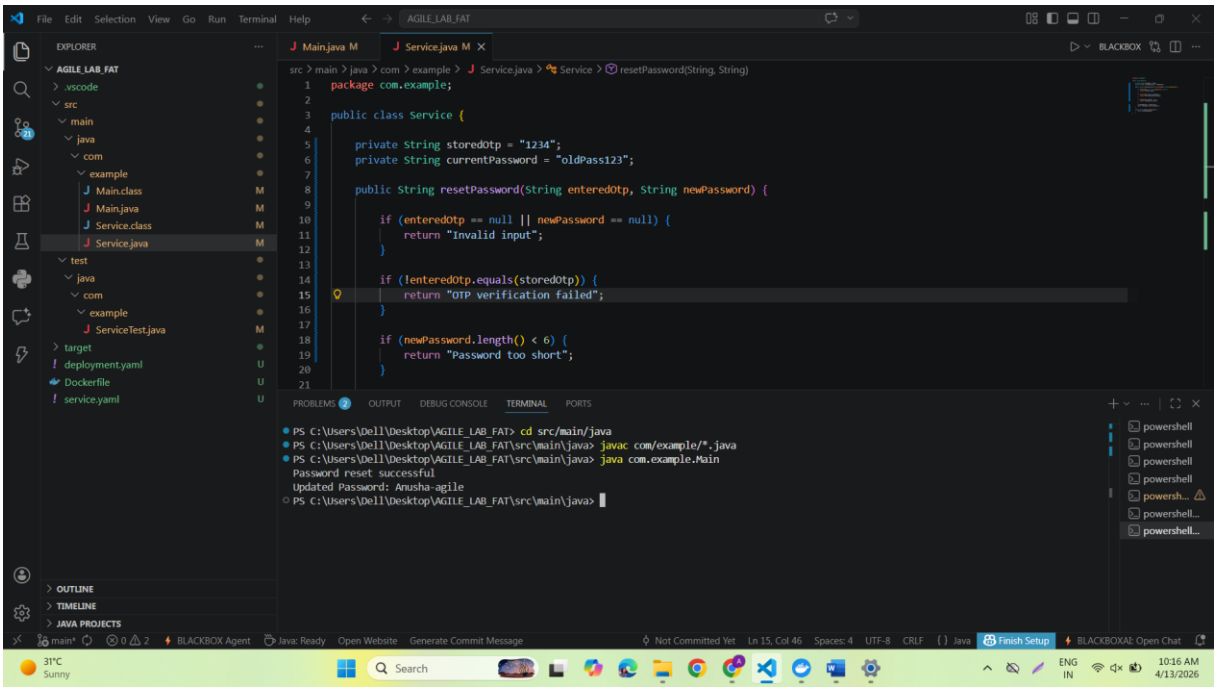
```java
package com.example;

public class Service {

    private String storedOtp = "1234";
    private String currentPassword = "oldPass123";

    public String resetPassword(String enteredOtp, String newPassword) {

        if (enteredOtp == null || newPassword == null) {
            return "Invalid input";
        }

        if (!enteredOtp.equals(storedOtp)) {
            return "OTP verification failed";
        }

        if (newPassword.length() < 6) {
            return "Password too short";
        }

        currentPassword = newPassword;
        return "Password reset successful";
    }

    public String getCurrentPassword() {
        return currentPassword;
    }
}
```

Run Java code
In terminal:

```
cd src/main/java
javac com/example/*.java
java com.example.Main
```

Expected output

```
Password reset successful
Updated Password: Anusha-agile
```




```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> cd src/main/java
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT\src\main\java> javac com/example/*.java
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT\src\main\java> java com.example.Main
Password reset successful
Updated Password: Anusha-agile
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT\src\main\java>
```

Step 3: JUnit Testing

pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>
    <groupId>com.example</groupId>
    <artifactId>agile-lab-fat</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>
        <dependency>
            <groupId>org.junit.jupiter</groupId>
            <artifactId>junit-jupiter</artifactId>
            <version>5.9.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>

    <build>
```

```
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.11.0</version>
                <configuration>
                    <source>17</source>
                    <target>17</target>
                </configuration>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>3.0.0-M5</version>
            </plugin>

            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.2.2</version>
                <configuration>
                    <archive>
                        <manifest>
                            <mainClass>com.example.Main</mainClass>
                        </manifest>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>

</project>
```

ServiceTest.java

```
package com.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class ServiceTest {

    Service service = new Service();

    @Test
    public void testInvalidOtp() {
        String result = service.resetPassword("9999", "newPass456");
        assertEquals("OTP verification failed", result);
    }

    @Test
```

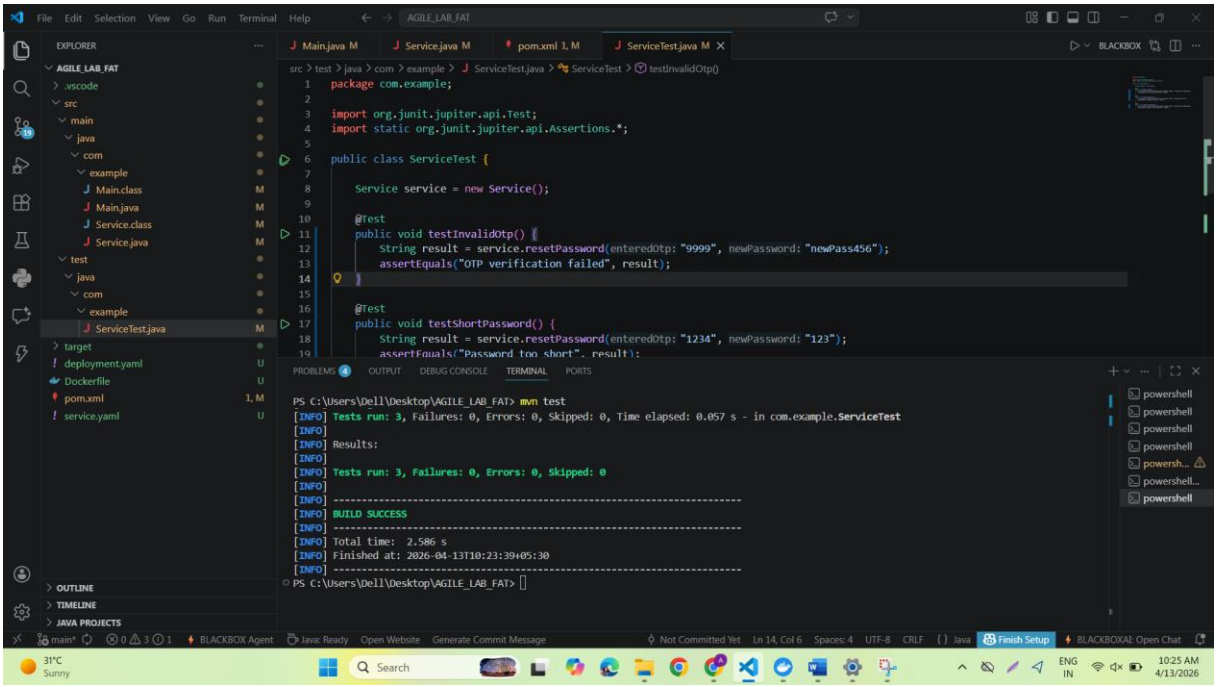
```java
    public void testShortPassword() {
        String result = service.resetPassword("1234", "123");
        assertEquals("Password too short", result);
    }

    @Test
    public void testSuccessfulReset() {
        String result = service.resetPassword("1234", "newPass456");
        assertEquals("Password reset successful", result);
    }
}
```

Run Java code
In terminal:

```
cd ../../..
mvn test
```

Expected output

```
BUILD SUCCESS
Tests run: 2, Failures: 0, Errors: 0
```



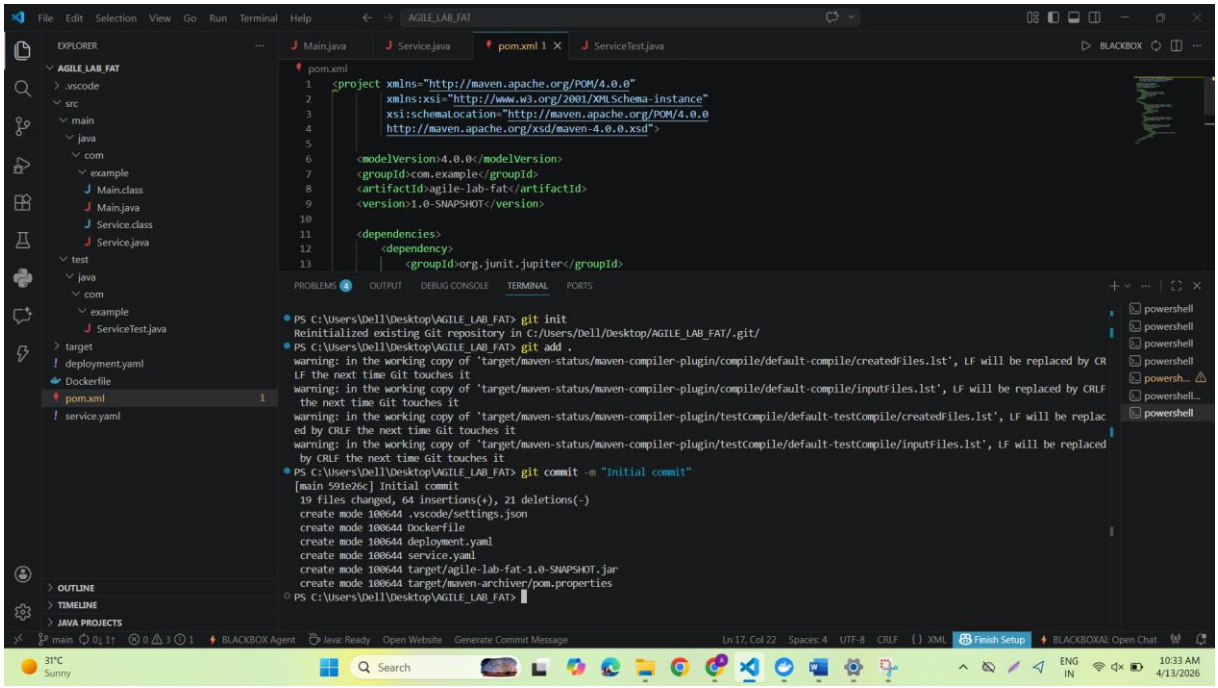
```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> mvn test
[INFO] Building agile-lab-fat 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ agile-lab-fat ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Dell\Desktop\AGILE_LAB_FAT\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ agile-lab-fat ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ agile-lab-fat ---
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Dell\Desktop\AGILE_LAB_FAT\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ agile-lab-fat ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- surefire:3.0.0-M5:test (default-test) @ agile-lab-fat ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.ServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.057 s - in com.example.ServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.586 s
[INFO] Finished at: 2026-04-13T10:23:39+05:30
[INFO] ------------------------------------------------------------------------
```

Step 4: Git + GitHub

1. Initialize Git

In terminal:

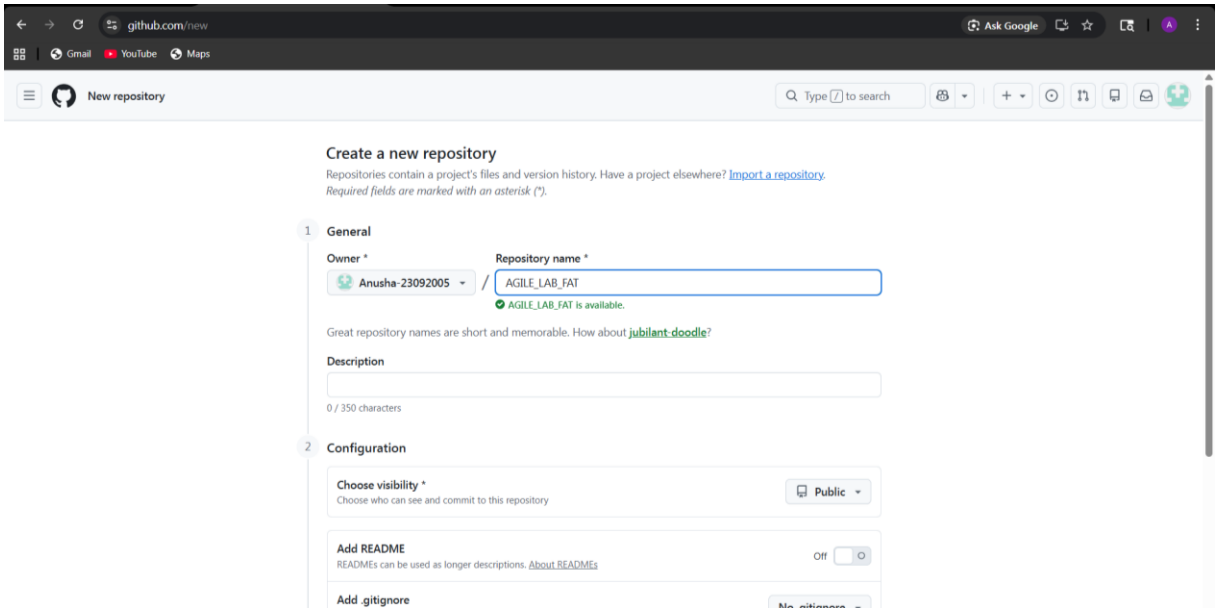
```
git init
git add .
git commit -m "Initial commit"
```

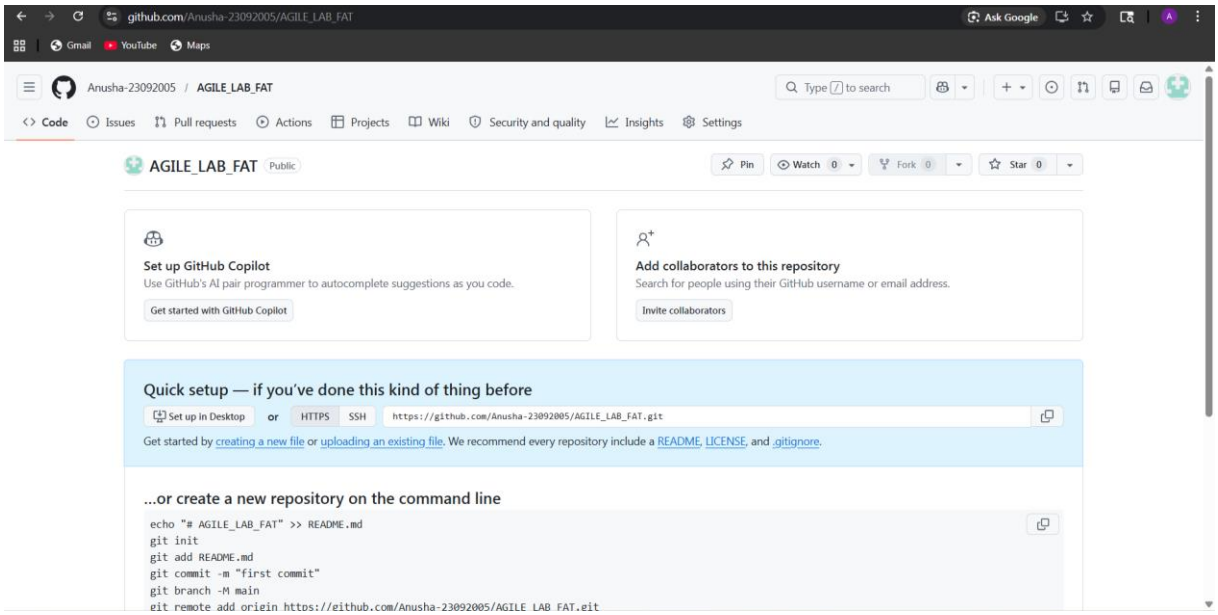



```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> git init
Reinitialized existing Git repository in C:/Users/Dell/Desktop/AGILE_LAB_FAT/.git/
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CR
LF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF
 the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replac
ed by CRLF the next time Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst', LF will be replaced
 by CRLF the next time Git touches it
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> git commit -m "Initial commit"
[main 591e26c] Initial commit
 19 files changed, 64 insertions(+), 21 deletions(-)
 create mode 100644 .vscode/settings.json
 create mode 100644 Dockerfile
 create mode 100644 deployment.yaml
 create mode 100644 service.yaml
 create mode 100644 target/agile-lab-fat-1.0-SNAPSHOT.jar
 create mode 100644 target/maven-archiver/pom.properties
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT>
```

2. Create GitHub repository

AGILE_LAB_FAT





3. Connect local project to GitHub

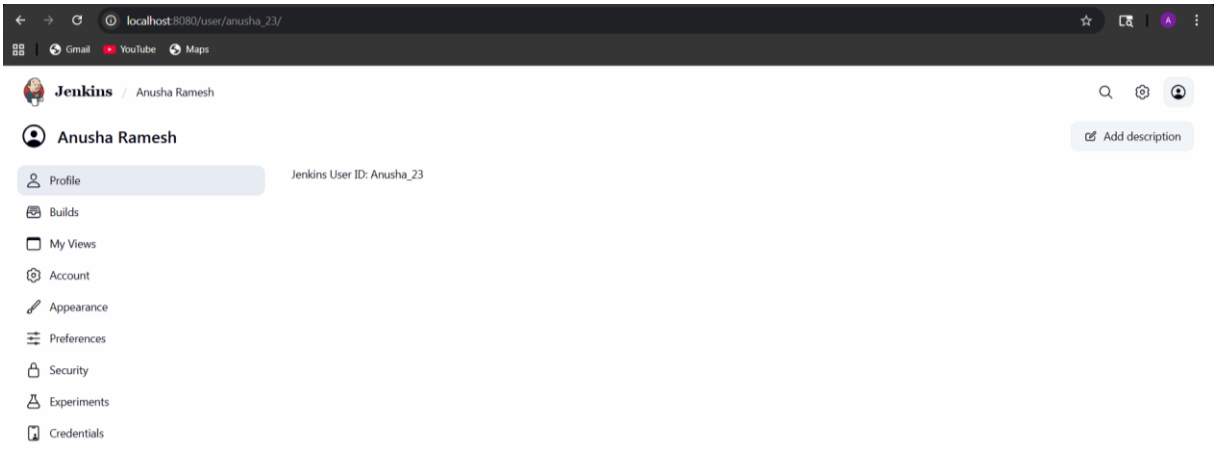
In terminal:

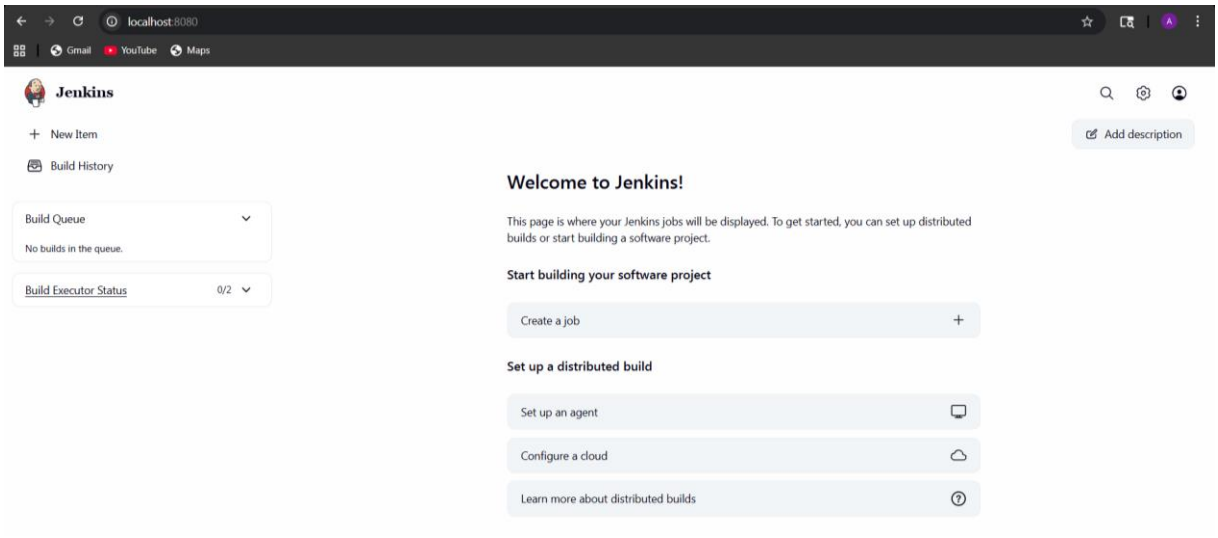
```
git branch -M main
git remote add origin https://github.com/Anusha-23092005/AGILE_LAB_FAT.git
git push -u origin main
```

```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> git branch -M main
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> git remote add origin https://github.com/Anusha-23092005/AGILE_LAB_FAT.git
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> git push -u origin main
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 8 threads
Compressing objects: 100% (23/23), done.
Writing objects: 100% (39/39), 6.32 KiB | 498.00 KiB/s, done.
Total 39 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/Anusha-23092005/AGILE_LAB_FAT.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT>
```

Step 5: Jenkins CI/CD pipeline

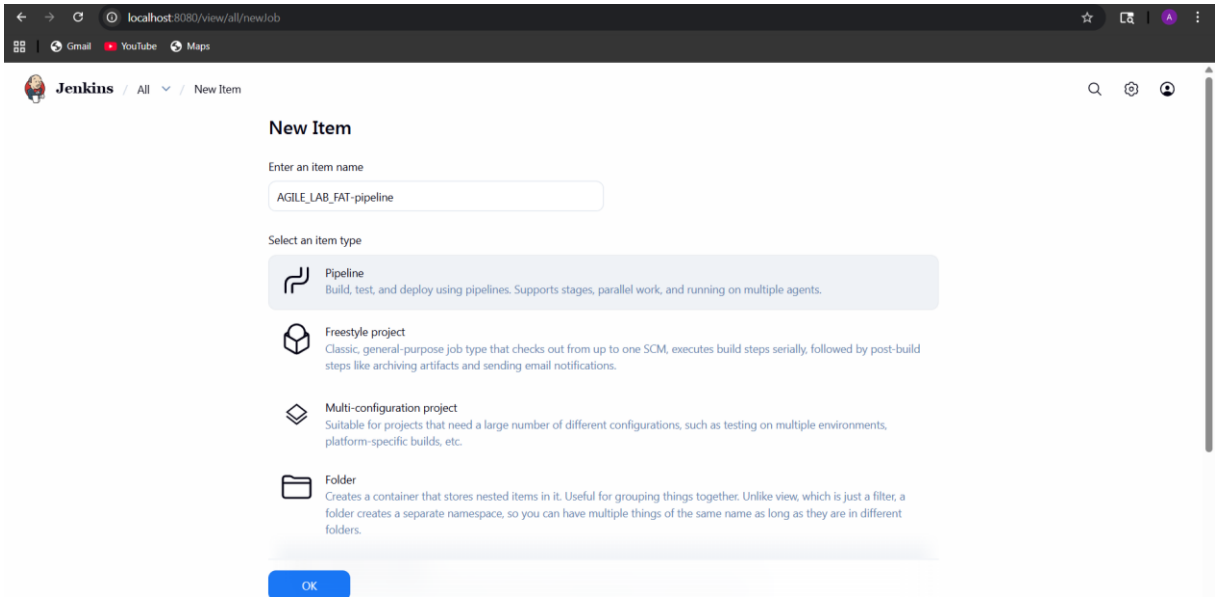
1. Open Jenkins: http://localhost:8080





2. Create a new Pipeline job

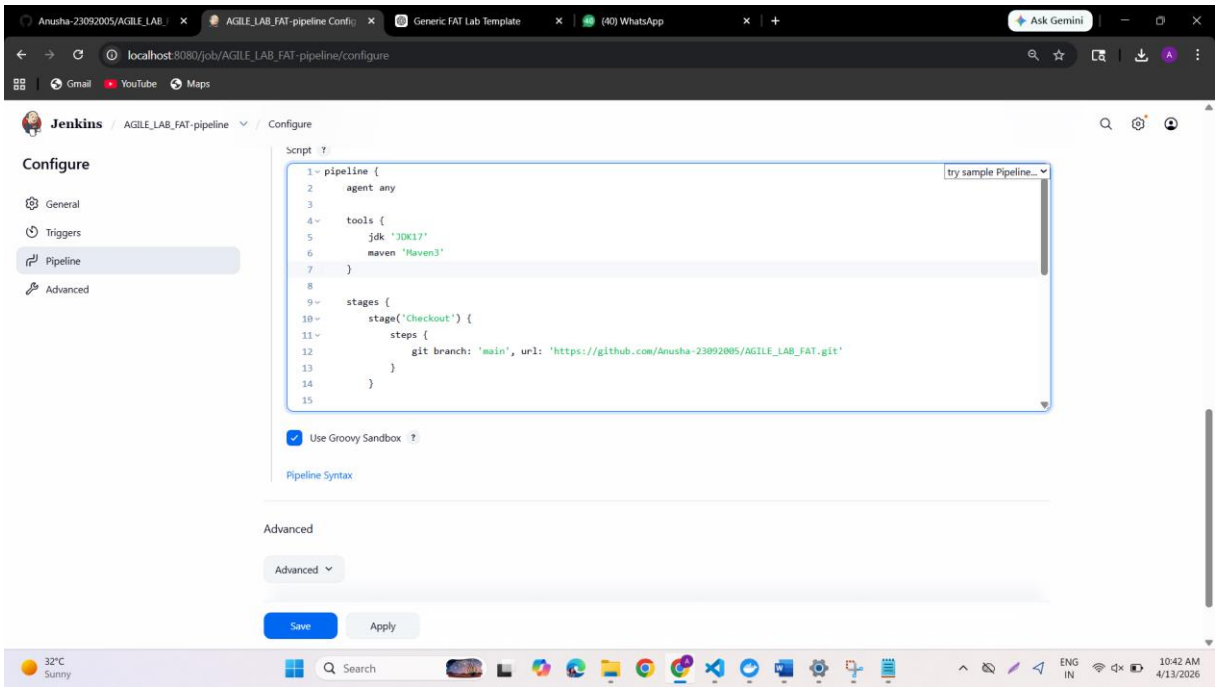
- Click New Item
- Enter name: AGILE_LAB_FAT-pipeline
- Select Pipeline
- Click OK



3. Configure GitHub repo in Jenkins

- Pipeline
- Pipeline script

```
pipeline {
    agent any

    tools {
        jdk 'JDK17'
        maven 'Maven3'
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/Anusha-
23092005/AGILE_LAB_FAT.git'
            }
        }

        stage('Build') {
            steps {
                bat 'mvn clean compile'
            }
        }

        stage('Test') {
            steps {
                bat 'mvn test'
            }
        }

        stage('Package') {
            steps {
                bat 'mvn clean package'
            }
        }
    }
}
```



4. Save and Build

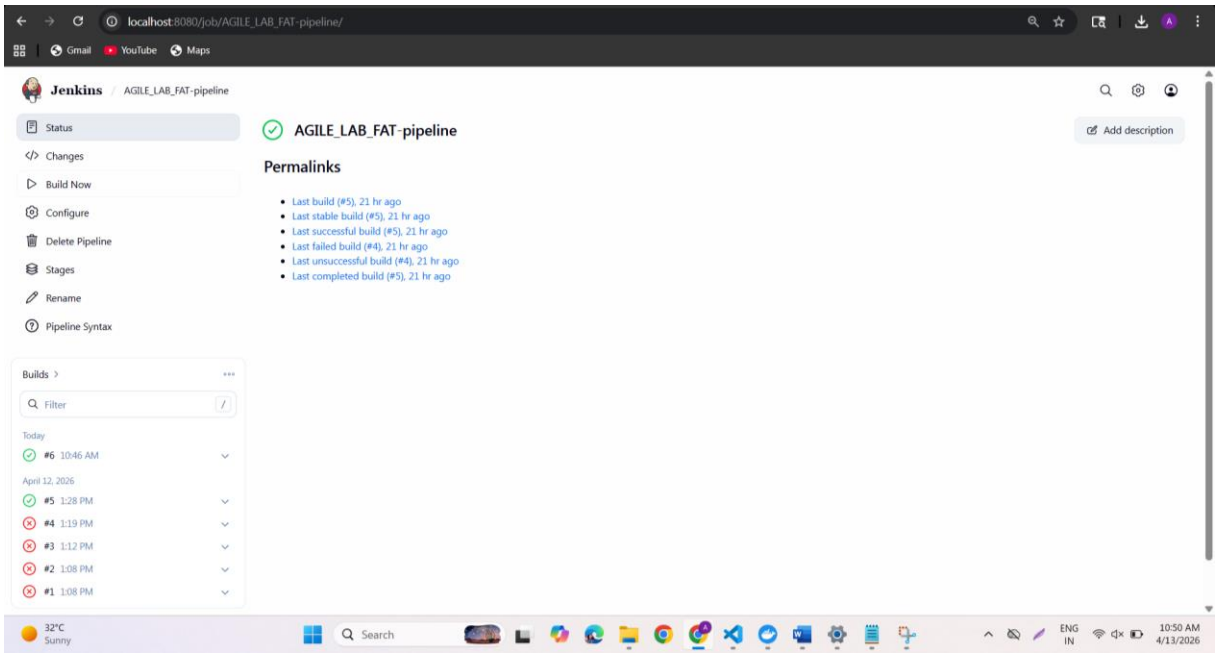
- Click Save

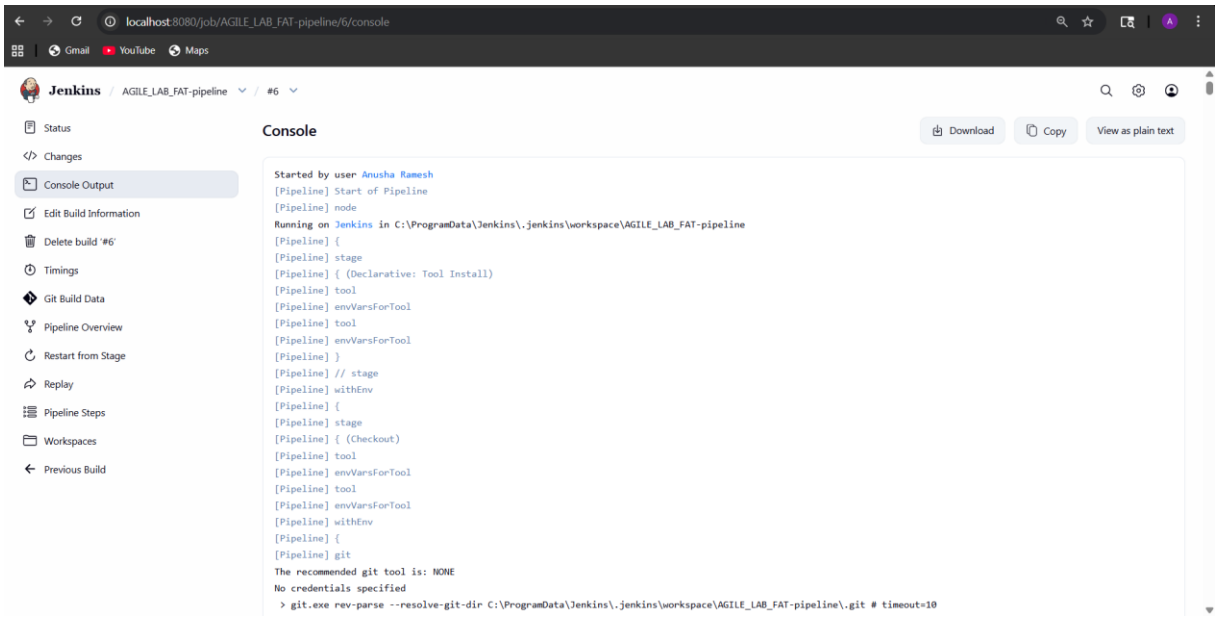
- Click Build Now

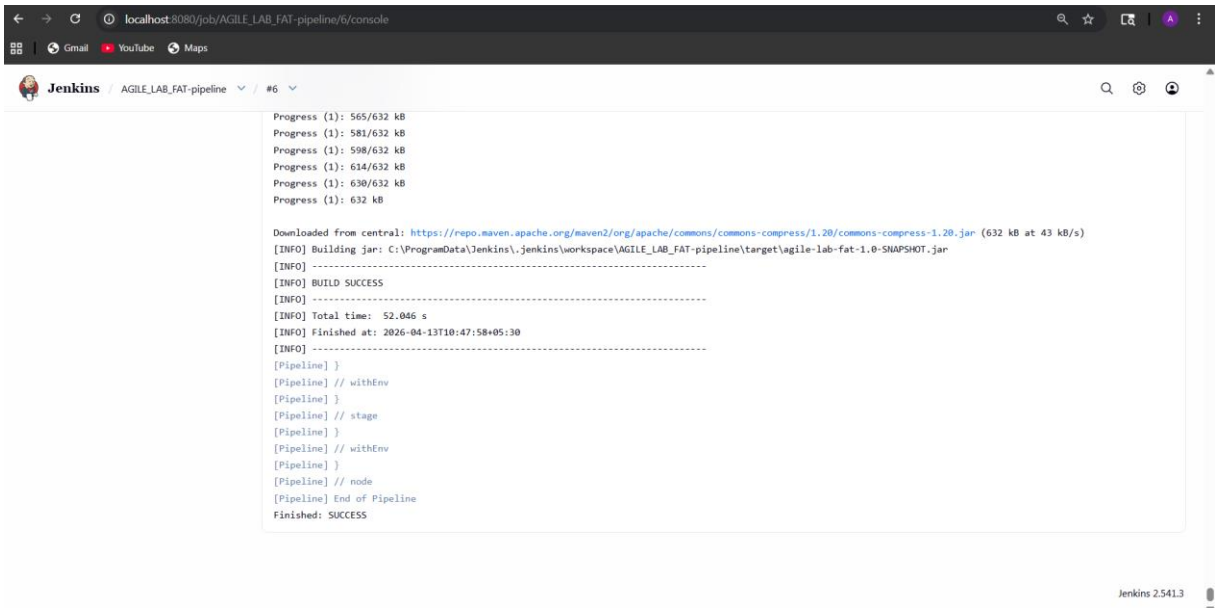
5. Check Console Output

Open build : Console Output

- checkout success
- build success
- test success
- package success


localhost:8080/job/AGILE_LAB_FAT-pipeline/
Jenkins / AGILE_LAB_FAT-pipeline
Status
Changes
Build Now
Configure
Delete Pipeline
Stages
Rename
Pipeline Syntax
AGILE_LAB_FAT-pipeline
Add description
Permalinks
Last build (#5), 21 hr ago
Last stable build (#5), 21 hr ago
Last successful build (#5), 21 hr ago
Last failed build (#4), 21 hr ago
Last unsuccessful build (#4), 21 hr ago
Last completed build (#5), 21 hr ago
Builds
Filter
Today
#6 10:46 AM
April 12, 2026
#5 1:28 PM
#4 1:19 PM
#3 1:12 PM
#2 1:08 PM
#1 1:08 PM


localhost:8080/job/AGILE_LAB_FAT-pipeline/6/console
Jenkins / AGILE_LAB_FAT-pipeline / #6
Status
Changes
Console Output
Edit Build Information
Delete build '#6'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Anusha Ramesh
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\AGILE_LAB_FAT-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\AGILE_LAB_FAT-pipeline\.git # timeout=10




```
C:\Users\Dell>sc query jenkins

SERVICE_NAME: jenkins
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE              : 4  RUNNING
                             (STOPPABLE, NOT_PAUSABLE, ACCEPTS_SHUTDOWN)
        WIN32_EXIT_CODE    : 0  (0x0)
        SERVICE_EXIT_CODE  : 0  (0x0)
        CHECKPOINT         : 0x0
        WAIT_HINT          : 0x0

C:\Users\Dell>
```

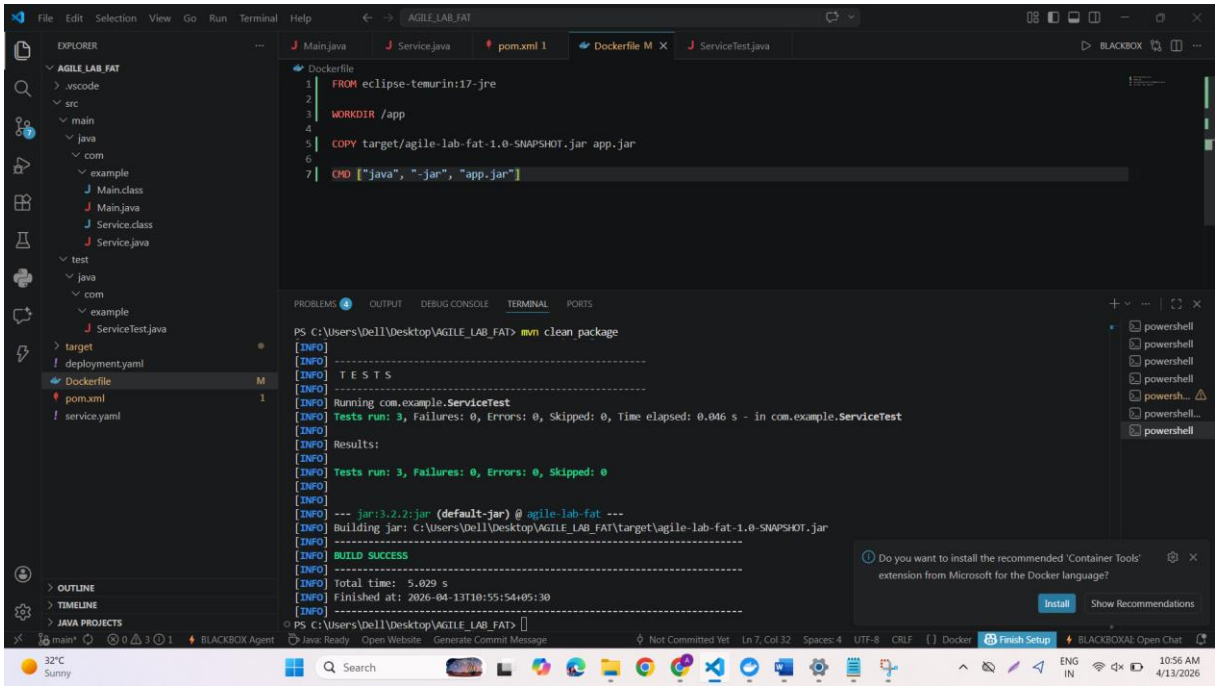
Step 6: Docker

1. Create Dockerfile

```
FROM eclipse-temurin:17-jre

WORKDIR /app

COPY target/agile-lab-fat-1.0-SNAPSHOT.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

2. Build JAR first

```
mvn clean package
```



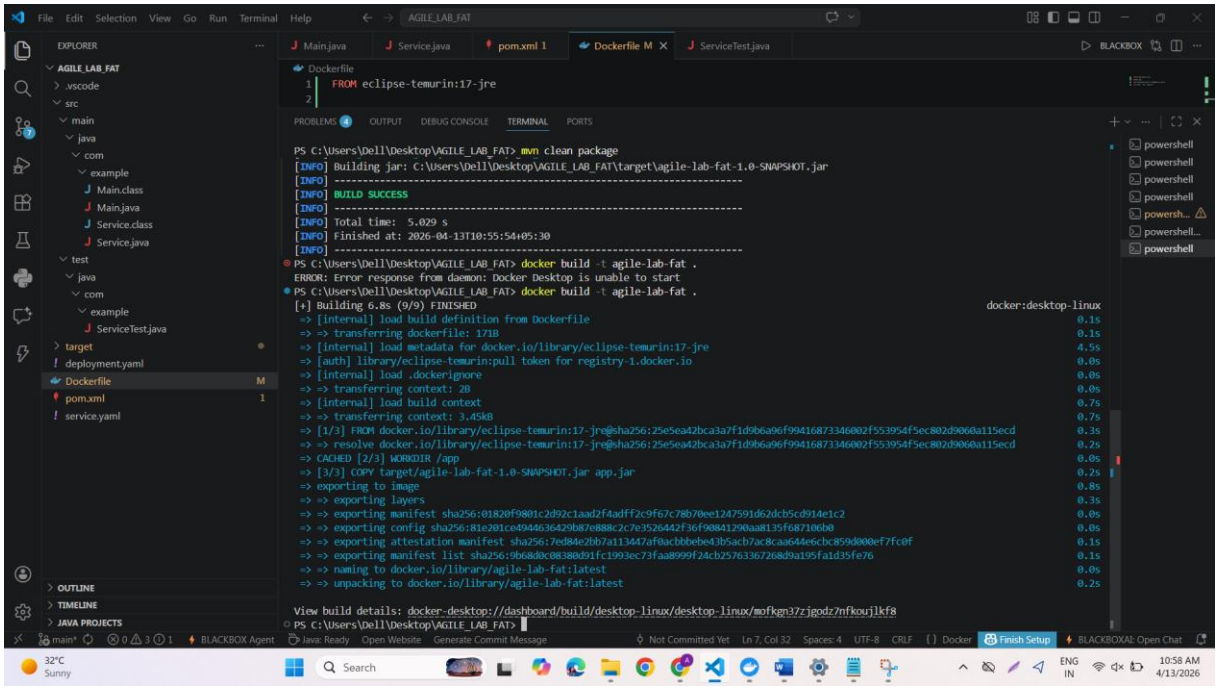
```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> mvn clean package
[WARNING] Using platform encoding (Cp1252 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\Dell\Desktop\AGILE_LAB_FAT\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ agile-lab-fat ---
[INFO] Changes detected - recompiling the module! :dependency
[WARNING] File encoding has not been set, using platform encoding Cp1252, i.e. build is platform dependent!
[INFO] Compiling 1 source file with javac [debug target 17] to target\test-classes
[INFO]
[INFO] --- surefire:3.0.0-M5:test (default-test) @ agile-lab-fat ---
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.ServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.046 s - in com.example.ServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.2.2:jar (default-jar) @ agile-lab-fat ---
[INFO] Building jar: C:\Users\Dell\Desktop\AGILE_LAB_FAT\target\agile-lab-fat-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.029 s
[INFO] Finished at: 2026-04-13T10:55:54+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT>
```

3. Build Docker image

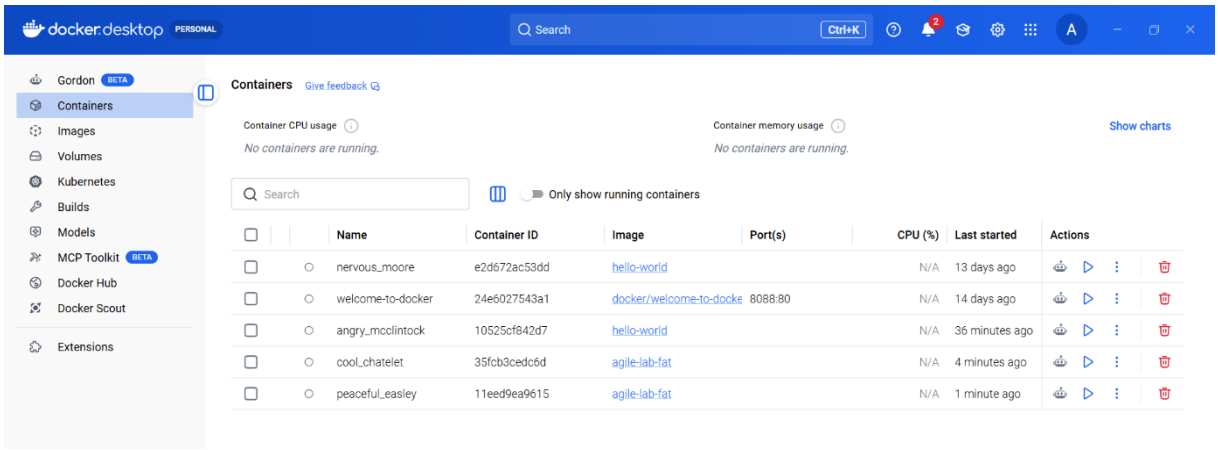
```
docker build -t agile-lab-fat .
```

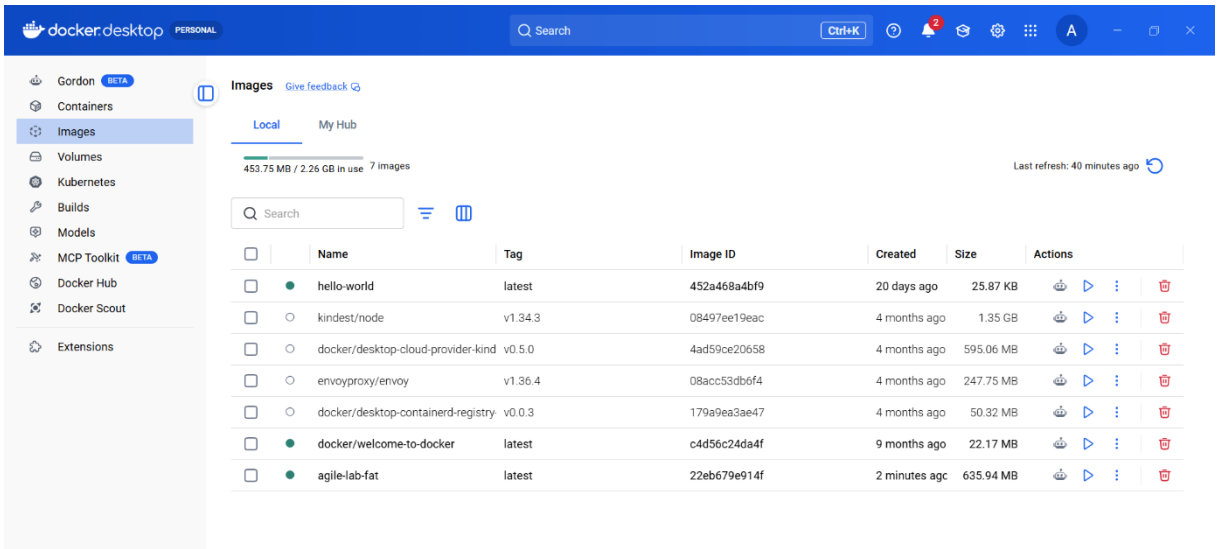
4. Run Docker container

```
docker run agile-lab-fat
```



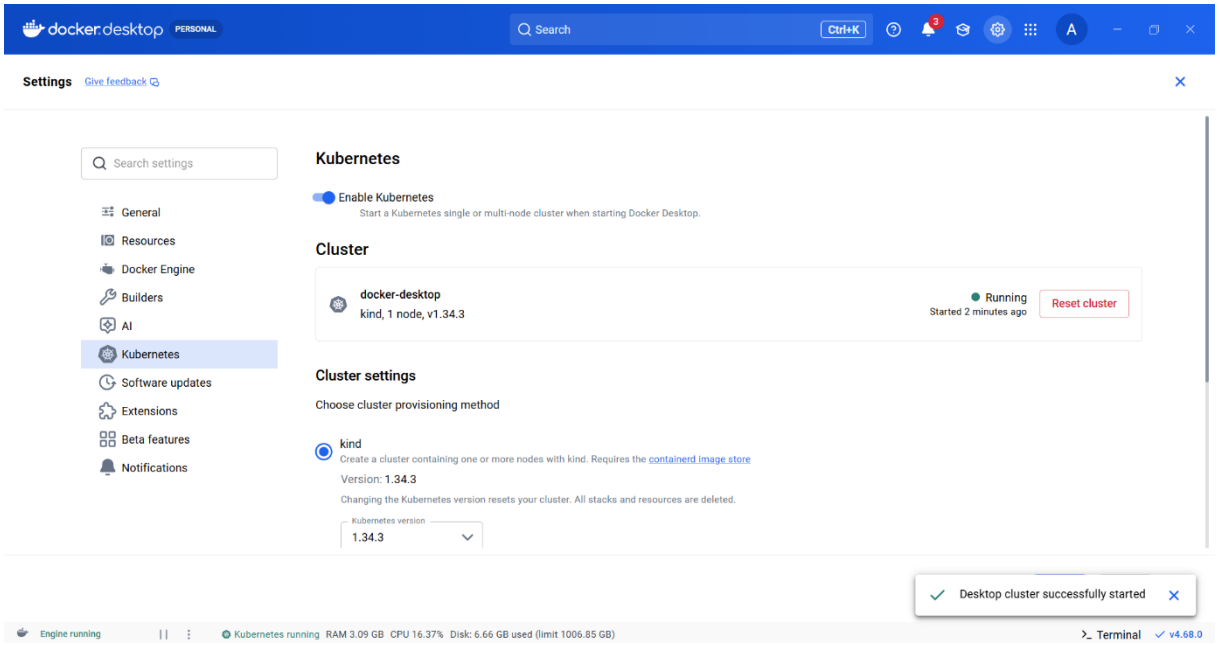
```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> docker run agile-lab-fat
password reset successful
Updated password: Anusha-agile
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT>
```





STEP 7: Kubernetes

1. Enable Kubernetes



2. Check using terminal

```
C:\Users\Dell>kubectl get nodes
NAME                   STATUS   ROLES           AGE    VERSION
desktop-control-plane  Ready    control-plane   3m45s  v1.34.3

C:\Users\Dell>
```

3. Create deployment file

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: agile-app
spec:
  replicas: 1
        selector:
          matchLabels:
            app: agile-app
        template:
          metadata:
            labels:
              app: agile-app
          spec:
            containers:
              - name: agile-container
                image: agile-lab-fat:v1
                imagePullPolicy: IfNotPresent
```

In terminal:

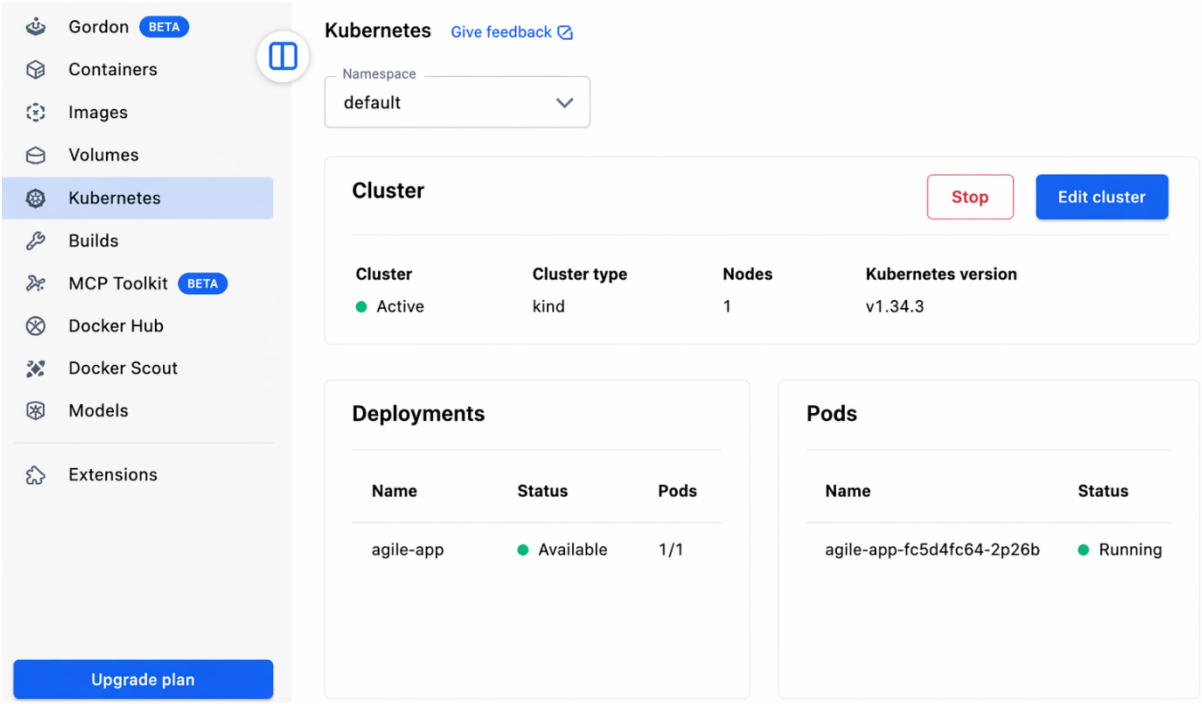
kubectl apply -f deployment.yaml

```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> kubectl apply -f deployment.yaml
deployment.apps/agile-app created
```

kubectl get deployments

kubectl get pods

```
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> kubectl get deployments
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
agile-app   1/1     1            1           14m
PS C:\Users\Dell\Desktop\AGILE_LAB_FAT> kubectl get pods
NAME                         READY   STATUS    RESTARTS   AGE
agile-app-fc5d4fc64-2p26b    1/1     Running   0          4m27s
```



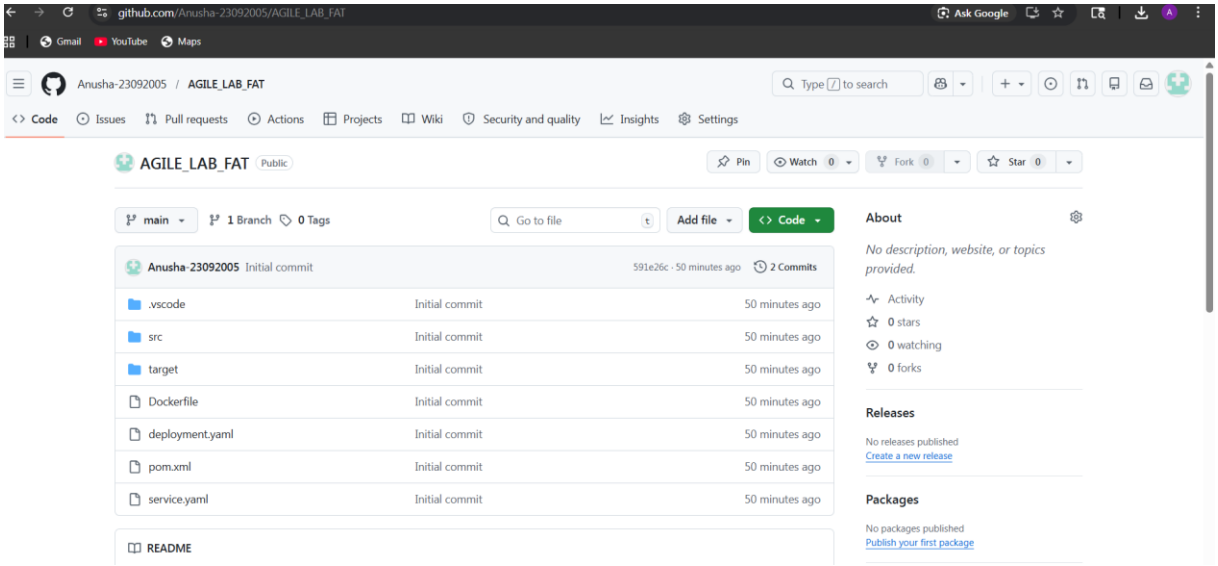
4.Create service file

```yaml
apiVersion: v1
kind: Service
metadata:
  name: agile-app-service
spec:
  selector:
    app: agile-app
  ports:
```

```
    - protocol: TCP
      port: 8080
      targetPort: 8080
  type: NodePort
```

In terminal:

```
kubectl apply -f service.yaml

kubectl get services

kubectl logs agile-app-68c8d858cc-stdlb
```



THANK YOU